Table 2: Mapping of symptoms and activity restrictions to health outcomes examined in epidemiological studies

	ARS[a]	RAD	ASD	REDV	RHA	CEDV	CHA
	Symptom Weightings (proportions)						
Upper Respiratory	0.70	0.40					
Eye irritation	0.30	0.15					
Asthma/COPD[b]		0.15	1.00	0.44	0.60		
Respiratory Infection		0.15		0.56	0.40		
Dysrhythmia		0.05				0.23	0.13
Congestive Heart Failure		0.05				0.27	0.28
Myocardial Infarction/Angina		0.05				0.50	0.59
	Activity Level Weightings (proportions)						
Mild limitations	0.90	0.80	0.90	0.74	0.50	0.41	0.34
Housebound	0.10	0.15	0.10	0.14	0.09	0.08	0.06
Need help caring for self		0.05		0.05	0.03	0.03	0.02
In hospital				0.07	0.37	0.48	0.57

[a]ARS = Acute Respiratory Symptom Day ASD = Asthma Symptom Day RAD = Restricted Activity Day REDV = Respiratory Emergency Department Visit RHA = Respiratory Hospital Admission CEDV = Cardiac Emergency Department Visit CHA = Cardiac Hospital Admission [b]chronic obstructive pulmonary disease

proportion of these less severe outcomes [1,22,37]. We therefore did not map these outcomes to the in-hospital level of activity restriction in generating our estimates.

Mapping of symptoms and levels of activity restrictions to a given outcome was derived from empirical data from our own emergency department study [29], and from a population-based survey in Strasbourg, France on the prevalence of minor respiratory symptoms [38]. Adjustments were made, where applicable, to shift the distribution towards the mild end of the spectrum to reflect those individuals who do not visit the emergency department and whose symptoms are presumably less severe on average.

The definitions of a number of health outcomes as examined in the air pollution epidemiology literature overlap with one another (e.g. hospital admissions and emergency department visits, restricted activity days and acute respiratory symptoms), and this is reflected in our mapping of these outcomes as shown in Table 2. This needs to be considered in order to avoid double counting of benefits among these health outcomes. Procedures have been suggested to deal with this, for instance by subtracting occurrences of restricted activity days, from those related to acute respiratory symptom days, prior to applying valuation estimates [1].

All inputs to our estimates pertaining to the duration of time in and out of hospital and of lost work days, and the probability of requiring inpatient care at the ward or critical care level (Table 3), were derived from Stieb et al. [29]. Based on data from the U.S. National Health Interview Survey, it was determined that work loss days comprise 21 percent of restricted activity days (standard error = 1.07%) [39] and we applied this factor in estimating lost productivity costs associated with restricted activity days. This is consistent with our mapping to the three levels of activity restriction, where 20% fell into the housebound or need help category. We assumed that 10 percent of asthma symptom days and acute respiratory symptom days were work loss days, based on the evidence described earlier.

Valuation estimates for cardiac and respiratory hospital admissions and emergency department visits were calculated as the weighted average among specific cardiac and respiratory conditions, with the weights based on the proportion of hospital admissions and emergency department visits accounted for by these conditions as reported in Burnett et al. [22] and Stieb et al. [33].

We obtained Monte Carlo estimates for each morbidity outcome by iteratively calculating its components (n = 1000), sampling from the observed distributions of model parameters, accounting for correlation among parameters, and from the observed distributions of input values. We report a central estimate which reflects model results evaluated at the mean of input values, and a 95% confidence interval bounded by the 2.5th and 97.5th percentiles of the output distribution. Where demographic information (age, education, income) and population health status were employed as independent variables in calculating

Table 3: Inputs on duration and disposition for valuation calculations

Condition	Admitted to:				Number of days:					
	Critical Care		Non-critical Care		In Hospital		Out of Hospital		Lost Work	
	Percent	Standard Error	Percent	Standard Error	Mean	Standard Error	Mean	Standard Error	Mean	Standard Error
	Individuals admitted to hospital									
Asthma	23.1%	4.8%	76.9%	4.8%	4.4	0.3	9.3	1.1	2.8	0.5
Chronic Obstructive Pulmonary Disease	0.0%	0.0%	100.0%	0.0%	7.9	0.7	9.4	2.6	0.8	0.6
Respiratory Infection	11.3%	3.4%	88.7%	3.4%	5.1	0.4	8.7	1.4	3.3	0.6
Congestive Heart Failure	12.9%	6.0%	87.1%	6.0%	7.2	0.7	8.9	2.4	1.2	0.6
Dysrhythmia	27.8%	10.6%	72.2%	10.6%	5.6	0.8	6.5	2.3	3.5	1.5
Myocardial infarction/angina	65.0%	3.8%	35.0%	3.8%	6.0	0.3	3.2	0.8	2.4	0.4
	Individuals visiting emergency department									
Asthma	2.6%	0.6%	8.5%	1.1%	0.6	0.1	12.9	0.5	1.2	0.1
Chronic Obstructive Pulmonary Disease	0.0%	0.0%	38.5%	5.5%	3.9	0.6	14.0	1.7	0.7	0.3
Respiratory Infection	1.5%	0.5%	12.1%	1.3%	0.9	0.1	14.3	0.5	1.5	0.1
Congestive Heart Failure	8.9%	4.2%	60.0%	7.3%	4.8	0.7	8.5	1.8	0.8	0.4
Dysrhythmia	9.6%	4.1%	25.0%	6.0%	1.9	0.5	5.3	1.0	1.7	0.6
Myocardial infarction/angina	61.8%	3.8%	33.3%	3.7%	5.8	0.3	3.1	0.8	2.3	0.4

stated-preference values, we conducted a sensitivity analysis by employing both the values of these variables in the original survey sample, as well as values based on the 1996 Canadian census and the 1996 National Population Health Survey [40]. While specific subgroups of those experiencing acute episodes of cardiorespiratory disease may be more specifically susceptible to air pollution as a trigger, and the severity of their disease episodes may differ from others, it is not possible to identify these groups with confidence. In order to deal with uncertainty generated by this issue, we also conducted a sensitivity analysis by varying the weight given to the least versus most severe levels of activity restriction and symptoms. When comparing our results with those from earlier studies, we converted values from other countries to the same year in Canadian dollars using purchasing power parity [41]. They were adjusted to 1997 Canadian dollars using the consumer price index for Canada for either all goods and services (WTP based estimates) or for health care (cost of treatment based estimates) [42]. While caution should be exercised in transferring our estimates to other countries, a similar procedure would be employed to do this, by applying the 1997 purchasing power parity between Canada and the target country, and adjusting to the currency year of interest using the country specific consumer price index for all goods and services.

Results

Cost of treatment model

Results from our analysis of cost of treatment data are presented in Table 4. The largest coefficients were associated with a diagnosis of congestive heart failure, and admission to critical care or non-critical care units. As reflected in the interaction terms, critical-care (CC) costs for respiratory infections (RI) and dysrhythmias (DYS) were considerably lower (RI*CC and DYS*CC), as were non critical-care (NCC) costs for asthma (AS*NCC). However, non critical-care costs were higher for those with a diagnosis of myocardial infarction/angina (MIA*NCC). Duration was associated with a coefficient of approximately $30 per day, but interaction terms essentially eliminated this effect for some diagnoses.

Comprehensive valuation estimates

Results of our calculations of overall valuation for each endpoint are presented in Table 5. Cost of treatment accounted for the majority of the overall value in the case of respiratory and cardiac hospital admissions, as well as cardiac emergency department visits. In the case of respiratory emergency department visits, cost of treatment accounted for approximately forty-five percent of the overall value. Lost productivity costs represented a small proportion of the overall value, although the proportion was greater for those endpoints which do not include a cost of treatment component. For both hospital admissions and emergency department visits, the overall value was larger for cardiac compared to respiratory conditions.

Table 4: Parameter estimates for cost of treatment model

Variable Name	Variable Description	Parameter Estimate ($)	Standard Error ($)
$V_{COT} = \alpha + \beta_{AS}AS + \beta_{CHF}CHF + \beta_{D}D + \beta_{CC}CC + \beta_{NCC}NCC + \beta_{AS*D}AS*D + \beta_{RI*D}RI*D + \beta_{MIA*D}MIA*D + \beta_{RI*CC}RI*CC + \beta_{DYS*CC}DYS*CC$			
Intercept	Intercept term	348.58	90.71
AS	Dummy variable for diagnosis of asthma (0,1)	440.33	125.97
CHF	Dummy variable for diagnosis of congestive heart failure (0,1)	1680.09	406.92
D	Total duration of disease episode (days)	31.70	7.85
CC	Dummy variable for admission to hospital in critical care unit (0,1)	4530.36	176.20
NCC	Dummy variable for admission to hospital in non-critical care unit (0,1)	1977.94	163.67
AS*D	Interaction term (see variable definitions above)	-27.71	9.17
RI*D	Interaction term where RI is a dummy variable for diagnosis of respiratory infection (0,1)	-29.02	8.07
MIA*D	Interaction term where MIA is a dummy variable for diagnosis of myocardial infarction/angina (0,1)	-30.54	14.50
RI*CC	Interaction term (see variable definitions above)	-1544.58	538.12
DYS*CC	Interaction term where DYS is a dummy variable for diagnosis of dysrhythmia (0,1)	-2402.73	906.97
AS*NCC	Interaction term (see variable definitions above)	-431.01	272.89
MIA*NCC	Interaction term (see variable definitions above)	1180.79	301.74
CHF*NCC	Interaction term (see variable definitions above)	-2158.82	539.88

This is primarily attributable to the larger cost of treatment for cardiac versus respiratory conditions. The ratio between total value and COI (i.e. cost of treatment plus lost productivity), ranged from 1.3 to 1.9 for those endpoints with a cost of treatment component, while the ratio tended to be larger for those endpoints without a cost of treatment component.

Results were not sensitive to choice of input values for demographic and health status characteristics (original survey sample characteristics versus nationally representative values for Canada). Sensitivity to the weighting given to the levels of activity restriction differed among the endpoints. The total value of avoiding respiratory and cardiac hospital admissions and emergency department visits varied by less than 10% from base estimates when we varied the weighting given to least versus most restrictive health states based on empirical data as described earlier. Relative to the base values, the total value of avoiding restricted activity days and asthma symptom days was respectively 54 and 57% lower and 65 and 104% higher when weighting of the different health states was varied. In all of the sensitivity analyses for restricted activity days and asthma symptom days, point estimates fell within the 95% confidence intervals on the base case estimates. However, the 95% confidence interval on the valuation estimate for asthma symptom days with less severe activity restrictions overlapped 0. For acute respiratory symptom days, valuation estimates for less and greater activity restrictions were respectively $12 and $28, both of which were within the 95% confidence interval on the base case estimate. In all cases the 95% confidence interval overlapped 0.

Discussion

Our estimates of the value of avoiding a range of acute effects of air pollution on cardiorespiratory morbidity are generally consistent with expectation in terms of the magnitude of values relative to severity and duration. The higher valuation of avoiding cardiac compared to respiratory disease episodes is driven mainly by the much higher proportion of cardiac conditions requiring admission to hospital, critical care, and invasive procedures [29]. Valuation for emergency department visits for both types of conditions includes cost of treatment related to hospital admission for those patients who were ultimately admitted. Because cardiac patients were much more likely to be admitted, this is reflected in a much larger cost of treatment value for cardiac compared to respiratory emergency department visits. While this would result in double counting of benefits where both hospital admissions and emergency visits are assessed and monetized in a benefits analysis, this approach was considered appropriate because concentration-response relationships for air pollution and emergency department visits are based on all those who visit the emergency department, including those who are ultimately admitted to hospital. Where both types of outcomes are being considered in benefits analysis, double counting can be avoided by using concentration-response relationships for emergency department visits to capture both types of effects, either by using relationships taken directly from epidemiological studies

Table 5: Valuation estimates[a] and ratio of total value to cost of illness, by endpoint and component

Endpoint	Cost of Treatment (CoT)	Lost Productivity (LP)	Pain, Suffering and Averting Expenditures		Total		Total Value/ (CoT + LP)
			In Hospital	Out of Hospital	Point Estimate	95% CI	
Respiratory Hospital Admission	$2,800	$300	$670	$410	$4,200	($3,400, $5,000)	1.3
Cardiac Hospital Admission	$3,800	$270	$760	$340	$5,200	($4,000, $6,400)	1.3
Respiratory Emergency Department Visit	$930	$160	$430	$520	$2,000	($1,700, $2,500)	1.9
Cardiac Emergency Department Visit	$3,200	$210	$680	$330	$4,400	($3,300, $5,600)	1.3
Restricted Activity Day		$25		$23	$48	($13, $82)	1.9
Asthma Symptom Day		$12		$16	$28	($11, $71)	2.3
Acute Respiratory Symptom Day		$12		$1	$13	($0, $28)	1.1

[a]estimates rounded to two significant figures

of air pollution and emergency department visits, or scaling up concentration response relationships from studies of hospital admissions, based on the expected ratio between hospital admissions and emergency department visits [1]. (See illustrative example below.)

Comparison with existing estimates

We present a comparison of our estimates with those from earlier studies [1–7,43,44] in figures 1,2,3 and Table 6 (see additional file 1). We have categorized estimates from earlier studies based on the authors' description of the health state, and the primary sources cited. It is difficult to make precise comparisons, however, because many other studies do not specify the level of activity limitation associated with the condition being valued, and our estimates pertaining to emergency department visits differ from earlier estimates in that they also capture costs for those patients who were ultimately admitted.

Nonetheless, in general, our estimates are comparable in magnitude to those from earlier studies in this area. *A priori*, we might have expected that our more comprehensive approach would have resulted in somewhat higher estimates compared to earlier studies. We observed that valuation estimates for the less severe outcomes were sensitive to specification of severity level, such that specifying a higher level of severity produced estimates at the higher end of those reported in earlier studies. It is also possible that in the context of the less severe outcomes which are associated with relatively low valuation, our approach, though more comprehensive, adds only small increments which do not result in a substantial change in magnitude of the overall estimate

Our estimates for hospital admissions are less than in most earlier studies. With respect to the cost of treatment component for this outcome, compared to the Anis et al. study [27], two studies based on U.S. data reported higher costs for pneumonia [45] and congestive heart failure [46], while another from Canada revealed similar costs for asthma hospital admissions [[47], personal communication, Dr. W. Ungar, November 2002]. Estimates from the U.K. were also similar to those derived here [6]. The estimates advanced by Chestnut et al. [1], though developed for Canada, are also based partially on U.S. data [48]. These findings may reflect true systematic differences in costs of treatment in the U.S. compared to Canada and the U.K.. It has been documented that intensity of care for some conditions is greater in the United States than Canada [49], and that administrative/overhead costs are significantly higher in the U.S. than in Canada in relation to the multiple versus single payer systems in the two countries [50,51]. Given that the ratios we observed for total value to COI were generally similar to (or indeed at the lower end of) those reported previously [1,4,15–17], this indicates that the component of valuation related to avoided pain and suffering and averting expenditures is also lower than earlier estimates. This may suggest that WTP to avoid health outcomes associated with significant health care costs may be lower in the context of a publicly funded universal health care system, possibly because of lower perceived risks of catastrophic financial consequences. Thus it appears that in the case of hospital admissions, the impact of our more comprehensive approach to valuation is overshadowed by lower costs of treatment in Canada compared to the U.S., which has been the predominant source of earlier such estimates.